UC DAVIS VETERINARY MEDICINE

Respiratory Disorders of Reptiles

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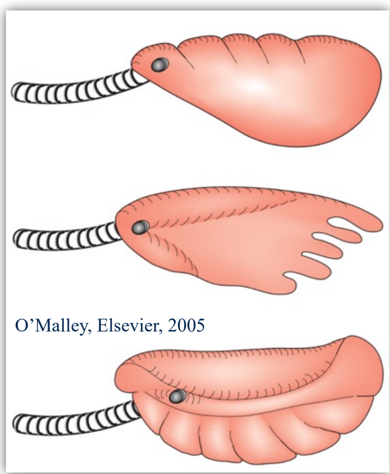
INTRODUCTION

Please refer to the lecture on the anatomy of reptiles for more background information on anatomy. Knowing the respiratory anatomy of reptiles is of great relevance when diagnosing and treating diseases of the respiratory system (see refresher below).

The most important features I would definitely recommend you brushing up on are:

- **Tracheal anatomy**: chelonians and crocodilians have **complete tracheal rings** (like birds) whereas squamates have **incomplete tracheal rings**. This has relevance when intubating these animals.

- **Pulmonary anatomy** is very variable with different types of lungs.



O'Malley, Elsevier, 2005

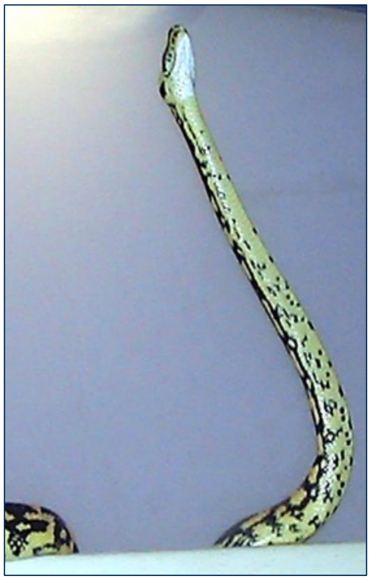
Most squamates have relatively simple lungs such as the **unicameral** (1 chamber) lungs of snakes and primitive lizards and the **paucicameral** (1 chamber too, but a bit more complex) lungs of most other squamates. Chelonians, advanced lizards (monitors essentially), and crocodilians have a more complex **multicameral** lungs. Most **snakes only have the right lung except primitive snakes such as boas and pythons.** In species with simple lungs, the caudal part of the lungs frequently consist of an **air sac** structure, particularly developed in snakes, skinks, and chameleons.

- **Coelomic cavities** are not homogeneous in reptiles as they are more diverse within their phylum than mammals or birds (which are technically avian reptiles). Regarding the respiratory system, you just need to remember that, in some species, the lungs are in a separate coelomic cavity (the pleural cavity) that is separated from the peritoneal cavity by a **post-pulmonary septum**. This is found in chelonians, crocodilians, and monitor lizards. This is important for surgery, coelioscopy, pulmonoscopy, and disease pathophysiology (less susceptible to ascites or mass compressing the lungs).

CLINICAL EVALUATION

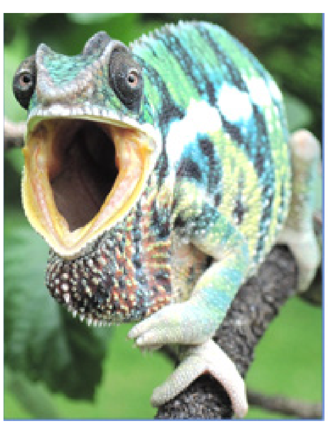
Well, reptiles with respiratory diseases do not necessarily present with respiratory signs. Most often, they are just **anorexic and lethargic** (like most reptile presentations). Upper respiratory signs are common with both upper and lower respiratory diseases. These include **bubbling** at the mouth or nostrils and nasal, oral, or tracheal **discharge**. In more severe cases, reptiles may present for open-mouth breathing, increased respiratory efforts and frequency, and displaying an orthopneic position. The orthopneic position depends on the species and the disease, but typically includes extension of the head.

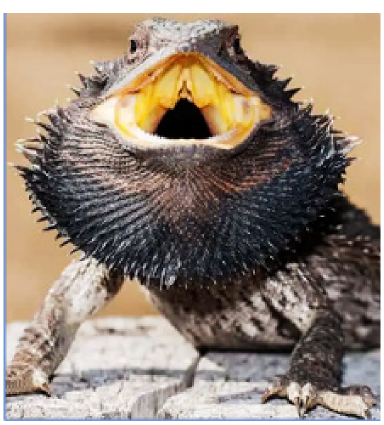
In snakes, the cranial third of the body may be held vertically.



Turtles may present with asymmetrical buoyancy when swimming with unilateral lung disease. Coelomic distention may also occur and can lead to dyspnea in species lacking a postpulmonary septum.

Before discussing the physical examination, it is important to be mindful of **normal species-specific reptile behaviors that could be misinterpreted for respiratory signs.** Many reptiles will adopt a defensive posture and display various behaviors when approached. These may include **hissing** (squamates), **gaping** (bearded dragons, many lizards), and **puffing up** (air in throat like in many lizards or in air sacs like in chameleons). **Gular pumping** may be a normal behavior in some chelonians. Also, most reptiles have **arrhythmic breathing with apneic pauses** (called apneustic breathing). Durring shedding, you may see pieces of skin around the nostrils, some being partially retained. In some species such as green iguanas, **nasal salt gland secretions** may be visible and should not be confused with nasal exudate.





Alright, now that this is out of the way, we can focus on the physical examination proper (as it relates to the respiratory system). On visual examination, pay a close attention to the **breathing pattern** such as the presence of an **orthopneic position, tachypnea, or increased respiratory efforts**. Rembember that dyspnea is less obvious in reptiles than in homeotherms. You may observe **nasal or oral discharge with or without bubbling**. This is also common with lower respiratory diseases as they frequently extend from a stomatitis and reptiles do not clear respiratory secretions well. **Periocular signs may also be present such as palpebral edema and conjunctivitis**, especially in chelonians. Again, aquatic turtles may have **buoyancy issues**.

Pulmonary auscultation is not very rewarding in reptiles and it is hard to hear anything. To avoid the friction of scales against the stethoscope, you can place a wet gauze first. You may hear stridor, crackles, and increased lung sounds.

Inspection of the oral cavity may be more rewarding as stomatitis is common with respiratory disorders, especially in snakes. Ptyalism, thick saliva, and various oral and tracheal discharges may be seen.

Coelomic palpation may detect a mass effect or the presence of ascites.

DIAGNOSTIC WORKUP

Blood Work

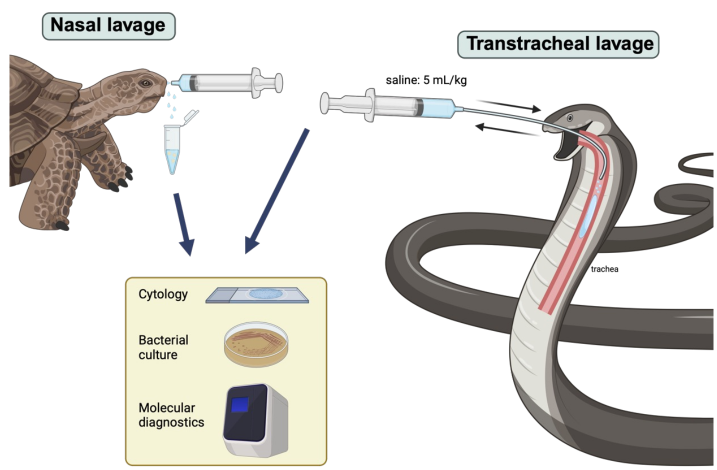
In terms of diagnostic tests, you should still start by the classic **CBC/biochem**. CBC and biochemistry findings are typically non-specific with most respiratory disorders. On occasion, you may observe an inflammatory leukogram or increased PCV with chronic respiratory disease. With inclusion body disease in boas, you may see inclusions in white blood cells on the blood smear. High bicarbonates on the biochemistry panel may be caused by respiratory alkalosis from the decreased ventilation. Many respiratory infectious diseases are also systemic in reptiles, so you may see evidence of that on the blood work.

Respiratory Lavages

Something that is very useful in squamates is **transtracheal lavage**, especially since many diseases are infectious. **This needs to be done sterily**, so the patient is typically anesthetized (lizards) or heavily sedated (snakes).

The glottis is cleaned and a sterile red rubber tube or shorter tube, depending on the species, is passed down the trachea as far as possible making sure not to contaminate the tip of the tube with oral secretions. Alternatively, the animal can be intubated first to ensure sterile passage to the trachea. Once in the trachea, about **5-10 ml/kg of sterile saline is administered**, then the animal is rocked side to side and tilted while fluid is reaspirated into the syringe. Gentle massage of the body may lead to more fluid being collected. The fluid can then be submitted for a variety of diagnostic tests such as **cytology, bacterial culture and sensitivity, fungal culture, and viral PCR depending on the differential diagnosis for that particular case**. Be aware that in snake, the culture and cytology may not necessarily be representative of lung pathology as the catheter is not in the lungs as they are too caudal.

In our lovely chelonians, we tend to do more **nasal lavage** than anything else because we tend to see them more for upper respiratory diseases. As they don't enjoy this as much as you would think, you need to deeply sedate them as well. Once sedated, the tortoise's head is pulled out while somebody else administer several ml of sterile saline into the nostrils using a small IV catheter while another person attempts to collect the dripping fluid into a sterile cup or tube.



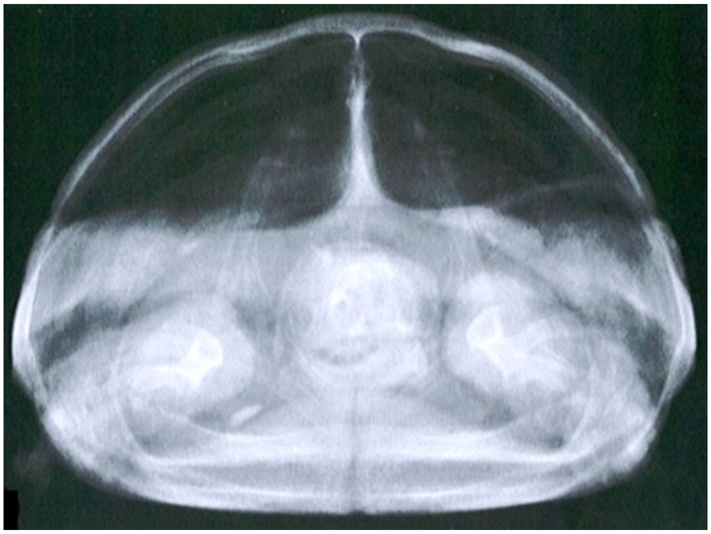
An alternative test is a **choanal swab,** but you need to be able to open the tortoise's beak for this, so they need to be more deeply sedated/anesthetized than for a nasal flush.

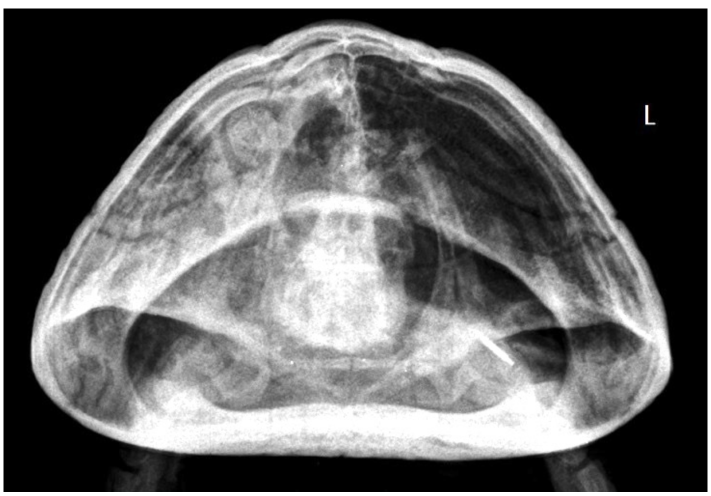
You can then submit this for a tortoise respiratory PCR panel.

Diagnostic Imaging

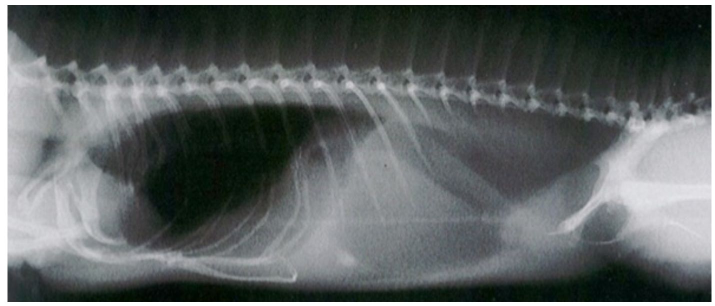
Radiographs are typically the first line of diagnostic testing that you should think about for respiratory disorders, especially for the lower respiratory system.

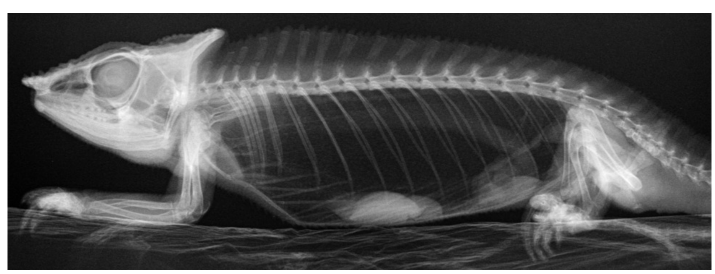
For chelonians, you need to take 3 views to have a good assessment of the lungs: **the DV, the lateral, and the craniocaudal**. You see the lungs well on the cranio-caudal and you can look for any asymmetry (see a normal and abnormal cranio-caudal xray views below).



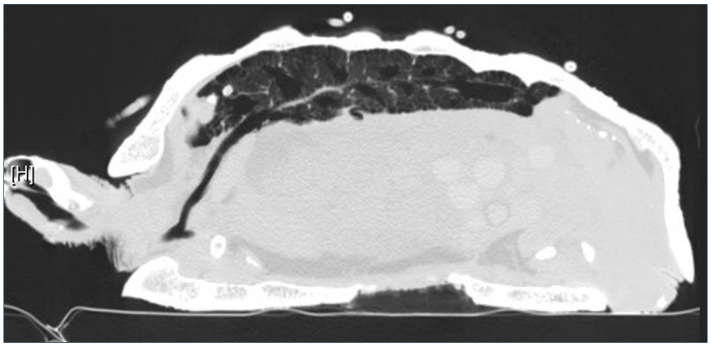


For lizards, you need to ensure you **position their thoracic limbs very cranially, away from the thorax,** especially if you want to look at the heart, which is very cranial in most reptiles. **For the lateral view, it is better in reptiles to obtain it in ventral recumbency, meaning you will need to rotate the beam.** This is because of their single pleuroperitoneal coelomic cavity, which may lead to fluid, masses, large organs moving over the lung when placed in lateral recumbency. See the correct positioning for an iguana and a chameleon below (note the chameleon is taped to the plate).



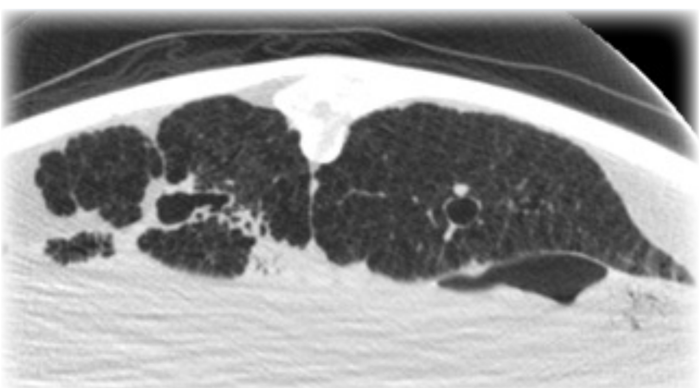


If you start seeing reptiles in your practice, you will quickly understand that xrays are not super useful in most cases because of the low contrast, especially in chelonians. For this reason, **computed tomography** will always be better if available and is strongly indicated in any chelonian. CT is particularly useful for respiratory diseases as you can see the nasal cavity and sinuses very well and you can differentiate the pulmonary parenchyma from air space. See below an example of a tortoise where you can very well see its multicameral lung.

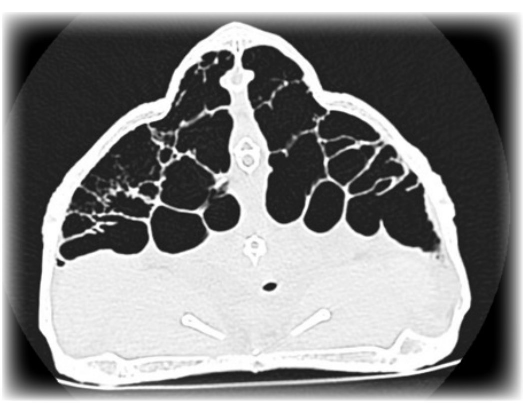


Below are some examples of respiratory diseases diagnosed on CT-scan.

Leopard tortoise with rhinitis



Green sea turtle with bacterial pneumonia and fibrotic lung

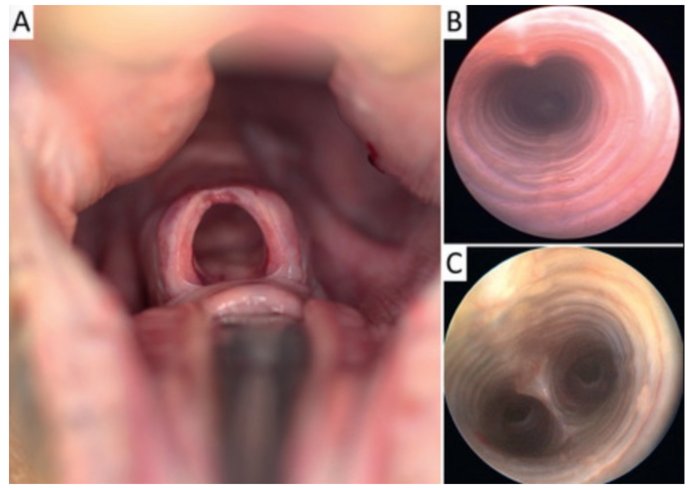


Leopard tortoise with emphysematous interstitial pulmonary fibrosis

MRI is also particularly useful in large tortoises, but is rarely performed due to cost.

Another very useful imaging technique is endoscopy. There are several endoscopic approaches available in reptiles and you'll see that they differ from other animals.

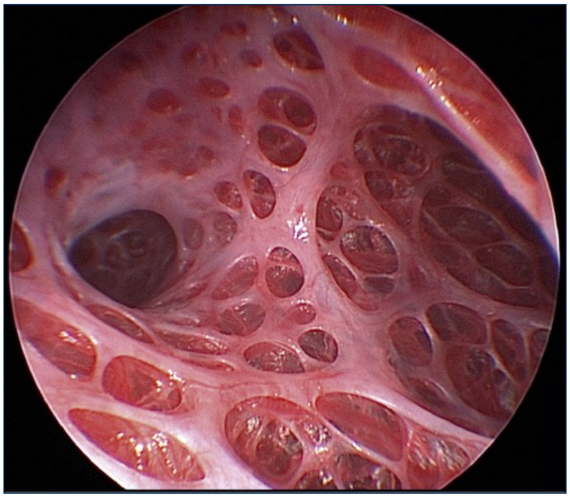
First, **tracheoscopy is seldom possible in many reptiles as the trachea is particularly small so it is only doable in large reptiles**. **The trachea of snake is also very long.**



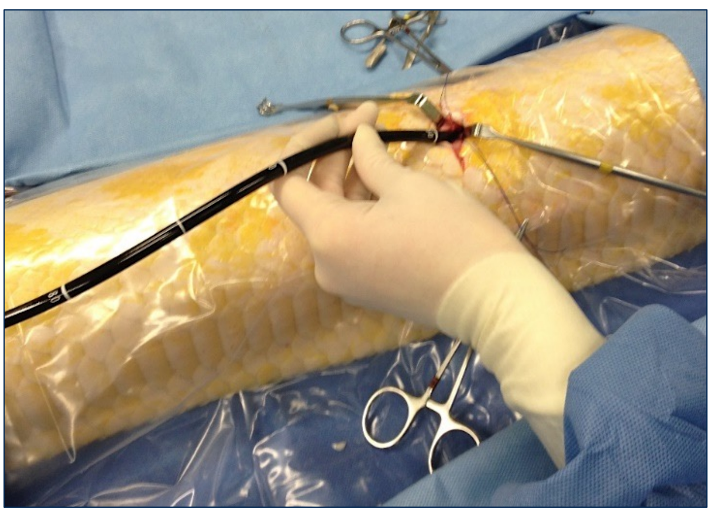
So going on the other end is more common and is known as **pulmonoscopy**. The pulmonoscopy approach is different depending of the species.

In chelonians, two approaches have been described, **transcarapacial** or through the **pre-femoral fossa**. For the transcarapacial, you will need to drill a hole through the carapace, preferentially after doing a CT-scan so you can target the diseased area. Then you can introduce the rigid endoscope, but you won't be able to move around that much, so it is mainly useful for sample collection.

For the pre-femoral fossa approach, you need to do a surgical pre-femoral coeliotomy first. Then the anesthetist gives a large breath allowing you to grab the post-pulmonary septum/ lungs. You make a hole through that and can place the endoscope. You will need to close that hole when done.

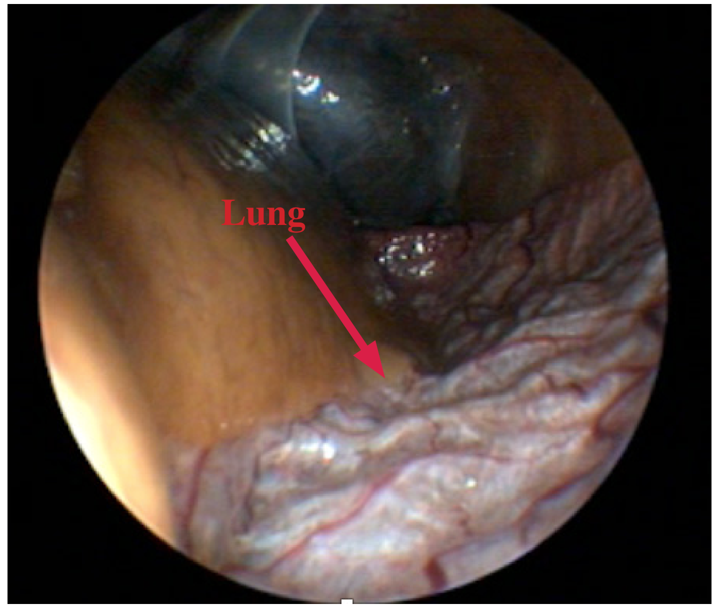


In snake, the procedure is easier. It needs to be done on the right for colubrids (they only have one lung) and on either side in boas and pythons. **An incision must be performed at about 35-45% of the distance from snout to vent so you end up in the avascular portion of the lungs**. You incise in an intercostal space, dissect the underlying tissue until you see the lung. You can grab it, place some stay sutures and make a stab incision into it. Insufflating the lungs can make it easier to identify the lung. You can use a flexible endoscope in large snakes or a rigid in smaller ones.



Once in there, you can examine the lungs, take some **samples for culture, and also some biopsies**. As all your samples are directly taken from diseased lung areas, your diagnostic tests will be more representative and reliable, however the procedure is much more invasive than a transtracheal wash. You can watch the procedure here.

Last, you can also simply **visualize the lungs during a routine coelioscopy as the lungs are not in a separate coelomic cavity in squamates**.



DISEASES

Generalities

There are many factors that may lead reptiles to have an increased susceptility to respiratory disorders. First, **poor husbandry,** as for any disease in reptiles, is a prominent factor that must be corrected. **A temperature that is too low or a humidity that is too low** are particular problems. For species requiring a high humidity, **adequate ventilation** of the enclosure should also be present.

It seems that the goal of many reptile breeders is to produce the most **inbred** pet reptile possible through the selection of recessive genes to generate interesting and more expensive color morphs. This in turn produces reptiles that are less healthy and more immunocompromised, and therefore more prone to respiratory infections.

The anatomy of reptiles also makes them prone to respiratory disorders. For instance, they have a **poorly developed respiratory mucociliary apparatus** making it harder for them to clear mucus and exudates. The lungs of many species (squamates) are also **free floating in the pleuroperitoneal cavity**, which makes them prone to being compressed by fluid, masses, and organomegaly. The **slow metabolism** of reptiles can make it difficult to resolve respiratory infection quickly. When treating reptiles for respiratory diseases, one may need to be mindful of their ability for **right-to-left cardiac shunting**, which may decrease pulmonary tissue drug levels for instance. Providing oxygen therapy may also be counterproductive in some species as it may decrease ventilation and increase shunting. Shunting is typically more pronounced in aquatic reptiles. On the plus side, it is possible to perform **intrapulmonary drug administration** in many reptiles through the placement of a catheter in most species or an air sac tube in squamates (particularly in snakes).

Most reptile respiratory diseases are actually infectious and **mixing species of reptiles from different taxonomic group or geographical areas may result in the spillover of infectious agents from one species, in which it is subclinical or less pathogenic to another species, in which it is associated with greater morbidity and mortality.** Good examples of that are viral diseases such as testudinid herpesviruses (Hermann's tortoises may die when exposed to Greek tortoises or Russian tortoises), inclusion body disease (pythons may die when exposed to boas, especially red-tailed boas), ferlavirus and many others. If you are interested in zoo medicine, this is a major hurdle to mixed species exhibits (not only for reptiles). The **snake mite** (*Ophionyssus natricis*) may mechanically vector some infectious agents from one snake to another as it has been demonstrated for *Aeromonas* spp. infection. **Overcrowding**, as seen in some pet shops or at some breeders or during transport, may also predispose reptiles to infections.

All in all, **chelonians have more upper respiratory diseases and squamates more lower respiratory diseases**.

Last, but not least, reptiles can function well in anaerobiosis for extended periods and may have a large respiratory reserve. As a consequence, **respiratory signs may only be noticeable when the disease is already very severe**, or clinical signs may be non-respiratory in earlier disease presentation (such as lethargy and anorexia).

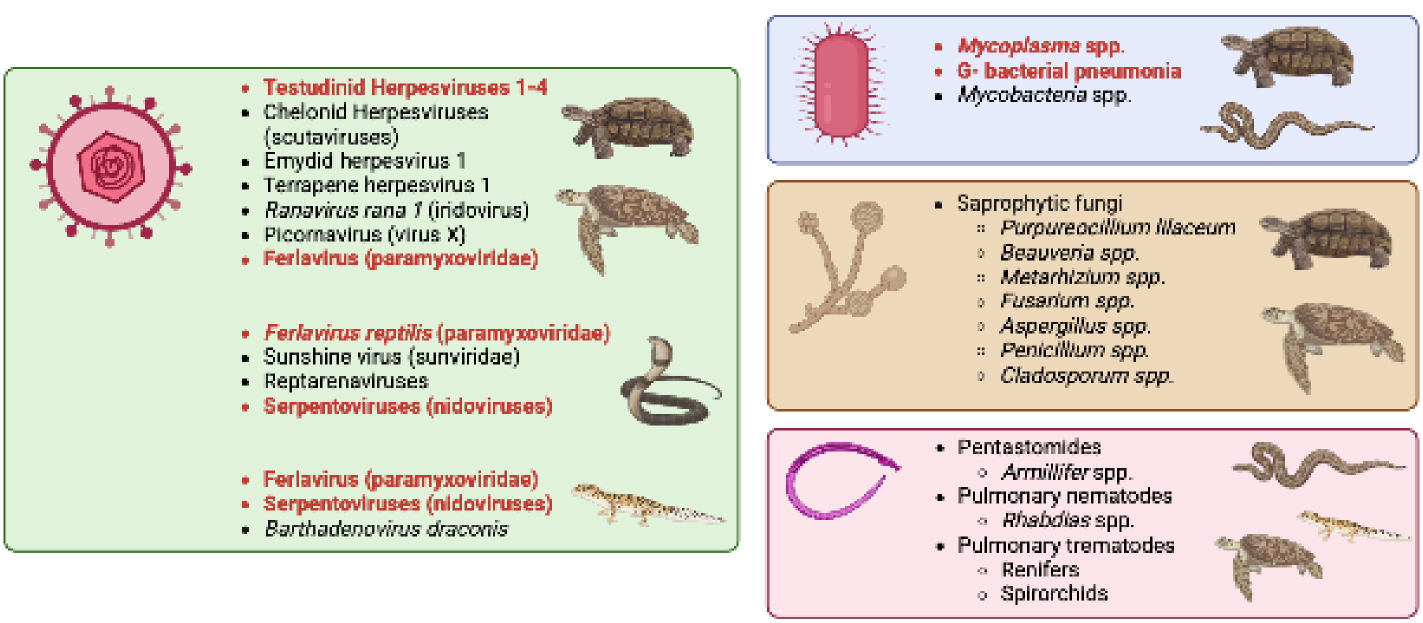
Infectious Diseases - Viruses

Most respiratory diseases of reptiles are actually infectious. Because there are a lot of species, there are a lot of infectious agents (see table below). Some of them are very important to know as they are very common.

Testudinid herpesviruses (genus *Scutavirus*) are respiratory and systemic infections of tortoises (family testudinidae). There are 4 different viruses, but only 2 are actually important (**TeHV-1 and 3**). As for all herpesviruses, **all survivors are permanently infected** and may start shedding again under a stressful event or intermittently. The big thing about TeHV is the species distribution and susceptibility. Most diseases seem to affect species of **Mediterannean tortoises**.

HV	Distribution	Hosts	Pathogenicity
TeHV-1	Eurasia	**Russian tortoise (carrier)** Greek tortoise Hermann's tortoise	Low mortality Hermann's tortoise very susceptible
TeHV-2	Americas	California desert tortoise	Low mortality
TeHV-3	Eurasia	**Greek tortoise (carrier)** Russian tortoise Hermann's tortoise Marginated tortoise	High mortality, widespread Hermann's and Russian tortoises very susceptible
TeHV-4	Africa	Angulate tortoise, leopard tortoise	Non clinical

In some species, the HV is less or not pathogenic, most likely because it has co-evolved with its host. But when this HV spillovers to another species, then more severe disease may ensue. For TeHV-1, it seems that Russian tortoises are mostly carriers, but other species can get seriously sick with it, in particular the Hermann's tor-



toise. For **TeHV-3, the most pathogenic of all**, the Greek tortoise seems to be the main carrier and other species get sicker, in particular Hermann's tortoises, again. In affected individuals, you may see **rhinitis, stomatitis, glossitis, conjunctivitis, tracheitis, all the way to systemic infection and hepatitis.** The big difference from mycoplasmosis in the upper respiratory signs is the presence of glossitis.

The shedding is through oronasal secretions. **The take-home message from this is that all survivors are permanently infected and you should not mix up species of Mediterranean tortoises.**

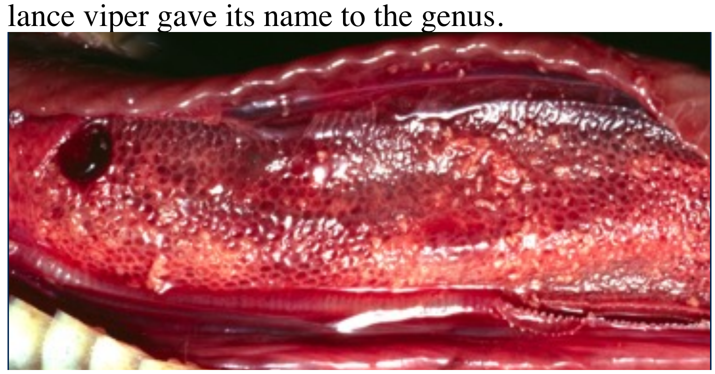
The diagnosis is primarily obtained through a **PCR on an oral/glossal swab**. Sensitivity is increased on a swab taken at the base of the tongue. PCR is particularly useful to identify clinical cases. Make sure the PCR is able to detect all TeHV. **Serology**, while not that useful for clinical cases, allow to identify carriers and previously exposed tortoises. There is no cross-reactivity between TeHV-1 and 3 on serology. On tissue, histopathology should be able to demonstrate **intranuclear inclusion bodies characteristic of herpesviruses**.

Treatment, if attempted, consists of supportive care, treatment of secondary infection, and the use of anti-herpesviral medications such as **acyclovir and ganciclovir**.

Turtles also have a wide variety of herpesviruses with respiratory tropism. **In freshwater turtles, two viruses are important: emydid herpesvirus 1**, which causes subcutaneous edema, nasal discharge, pneumonia, and systemic disease in Western pond turtles (our only native freshwater pond-type turtle), painted turtles and map turtles; **and terrapene herpesvirus 1**, which causes stomatitis, rhinitis, tracheobronchitis, pneumonia and systemic disease in Eastern box turtles.

In sea turtles, there are also a number of herpesviruses of importance. Three viruses cause respiratory lesions, namely the **lung, eye, trachea disease (LETD) in green sea turtle, the loggerhead genital-respiratory virus (LGRV), and the loggerhead orocutaneous herpesvirus (LOCV) both in loggerhead sea turtles.** The LETD can cause gasping, harsh respiratory sounds, buoyancy abnormalities, and inability to diver properly. The LOCV can also cause pneumonia. There are two other herpesviruses in green sea turtles that seldom cause respiratory lesions, those are the gray patch disease and fibropapillomatosis (chelonid herpesvirus 5).

Iridovirus (*Ranavirus rana 1, also known as frog virus 3*) is an envelopped DNA virus of ectothermic animals that is not very species-specific and can infect fish, frog, and reptiles. **It is mainly an issue in amphibians, but is seen on occasion in freshwater turtles, box turtles, and tortoises.** In freshwater turtles, it causes subcutaneous edema (known as "redneck disease"), nasal discharge, conjunctivitis, stomatitis, pneumonia, hepatitis, and systemic disease. The diagnosis is, like for most of these viruses, with a PCR on oral and cloacal swab or on tissue samples. Let's move on to snakes with **ferlavirus**. *Ferlavirus reptilis* is a species of paramyxoviridae that causes **neurorespiratory disease**. It is a very important disease of snakes, but it can also infect chelonians and lizards. **It can be seen in most snakes such as colubrids, boids, and pythonids, but is especially pathogenic in venomous snakes of the viperidae (vipers, rattlesnakes) and elapidae (cobras, mambas) families**. The fer-de-lance viper gave its name to the genus.



The virus has a tropism for the neurologic and respiratory systems and infection may lead to dyspnea, oral exudate, head tremors, opisthotonos, and death. On necropsy, pulmonary congestion and hemorrhage are seen. **The diagnosis is made by PCR on tracheal swabs, transtracheal wash, or tissue samples.**

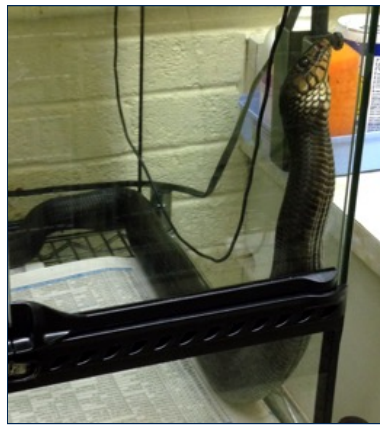
Another virus causing similar signs in Australian pythons is the **sunshine virus** (family of sunviridae), but it is very rare outside of Australia.

The second viral disease of great important in snakes is **nidovirus, also known as serpentoviruses**. These envelopped RNA viruses were discovered in **ball pythons and were found to be one of the most common causes for pneumonia and respiratory disease in this species**. So mainly an infection of pythons, but nidoviruses can be seen in boids and colubrids too and there are some rare reports in lizards (veiled chameleons, shingleback lizard) and chelonians (snapping turtles). The **prevalence is high in captive pythons**. The disease is characterized by ptyalism, stomatitis, wheezing, esophagitis, dyspnea, and death. Transmission is via respiratory fomites, fecal-oral, or direct contact.

The diagnosis can be made by PCR on oral swab, transtracheal wash, or lung tissue samples. It seems that non-clinical shedders are possible too.

Infectious Diseases - Bacteria

Pneumonia is very common in snakes, in particular in pythons (especially in Burmese pythons) and boas. While serpentoviruses play a big role in many of these infections, primary or secondary (to serpentovirus) bacterial pneumonia are also common. Do not forget ferlavirus as a differential here too. A **suboptimal husbandry,** in particular temperature, humidity, and ventilation plays a big role here. **Samples for culture can be obtained with a transtracheal wash or lung sampling by pulmonoscopy**. Typically a mix of **gram-negative bacteria** are grown. The treatment may be challenging and **antibiotic treatment for extended periods of time is often necessary. Ceftazidime** is frequently given in snakes for this as an injection lasts 3 days and the antibiotic covers most bacteria typically recovered from these infections. You should also improve the husbandry and you can add some antibiotic nebulization. See below some snakes with pneumonia, one with the first third of the body held upright (orthopneic position in snakes) and one with open-mouth breathing and lots of oral mucus.



Mycoplasmosis of tortoises is very common and an important disease to know. While many tortoises are susceptible and may get infected with *Mycoplasma* spp., it is **particular a problem in the California desert tortoise (our official state reptile) and the gopher tortoise, both in captivity and in the wild**. The prevalence can be pretty high and the infection has played a significant role in the population decline in these 2 species.

It is caused mainly by ***Mycoplasma agassizzi***, but also to a lesser extent by ***Mycoplasma testudineum***.

The disease is characterized by **high morbidity and low mortality. Nasal discharge, rhinitis (erosive over time), conjunctivitis**, and ultimately pneumonia are seen. The big difference with testudinid herpesviruses, is that there is **no stomatitis or glossitis** with mycoplasmosis. **The infection is also thought to be chronic and tortoises seem to be permanently infected. The diagnosis is made by a PCR of a nasal flush or a choanal swab.** As for diseases potentially caused by different agents, you need to know what you are testing with the PCR. The PCR can be a specific *M. agassizzi* PCR or a pan-mycoplasmal PCR depending on the lab.

Because there are many carriers and permanently infected animals, **serology plays an important role in screening tortoises**, especially in conservation programs. Culture is typically not performed for clinical patients as *Mycoplasma* spp. are fastidious growers and may take over 6 weeks to grow.

Treatment is actually difficult as you can never be sure that these tortoises are completely cleared of the *Mycoplasma*, they should be considered permanently infected. Antibiotics with intracellular penetration should be chosen. **Macrolides are the drug of choice such as clarithromycin, tulathromycin, and azithromycin**. Topical eye drops with antiinflammatories and antibiotics may also be used.

Infectious Diseases - Others

You are not required to know these agents, but these are part of the differential diagnosis of respiratory diseases in reptiles, depending on the species.

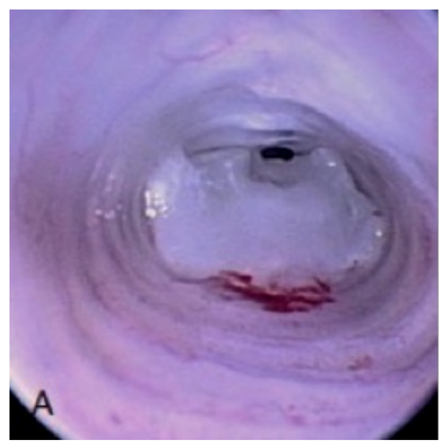
Fungal pneumonia is seen on occasion in reptiles. It seems to be particularly common in **cold chelonians, especially in giant tortoises and cold-stunned sea turtles.** A variety of fungal species have been isolated, most of them being saprophytic opportunistic pathogens. The two most important, as far as respiratory infections are concerned, are ***Purpureocillium lilaceum* and *Beauveria bassiana***, but many others have been isolated (see previous table). *Metarhizium* spp. are also commonly encountered in chameleons, affected the tongue and oral cavity.

For parasites, we have some interesting species in reptiles. **Pentastomides** are ancient worm-like crustaceans that are mainly seen in wild-caught snakes and crocodilians. They can actually infect humans as intermediate host. They may lead to obstruction of airways in snakes and they have to be removed manually or endoscopically. Most of the time, they are minimally pathogenic. Pulmonary nematodes include ***Rhabdias*** spp. also known as lung worms of reptiles. They penetrate the skin or they are ingested and then migrate to the lungs from the GI. **Renifers and spirorchids** are trematodes. Renifers are mainly seen in snakes and are mostly non-clinical. You can spot them in the oral cavity, but they can also migrate to the lungs. Spirorchids are mainly common in aquatic turtles including freshwater and sea turtles. They are intravascular parasites that can lead to pneumonia. Intermediate hosts are snails.

Non-Infectious Diseases

There is no particular non-infectious disease that is very common and a wide variety of non-infectious diseases may affect the respiratory system in reptiles.

In terms of neoplasia, **lymphoma** is diagnosed with some frequency in reptiles. One particular neoplasia of note is **tracheal chondroma of ball pythons** that may lead to tracheal obstruction. The treatment is tracheal resection and anastomosis. An **air sac tube** may also be placed pending surgery.



Fibropapilloma of sea turtles may lead to internal masses, some of them being in the lungs. Other neoplasias less commonly reported in the respiratory system of reptiles include fibroma, fibrosarcoma, fibroadenoma, adenocarcinoma, squamous cell carcinoma, plasma cell tumors, and metastasis of other tumors.

Other reported non-infectious diseases include **trauma** with penetrating injuries to the lungs (chelonians being ran over, lawn mowers, boat for sea turtles) or head injuries affecting the nasal cavities. **Respiratory foreign bodies** may include insect body parts in insectivorous reptiles, plant materials in upper airways in chelonians and fish hook in aquatic chelonians. **Dysecdysis** may lead to increased respiratory sounds by blocked nares by unshed skin. **Hypovitaminosis A** in turtles may lead to upper respiratory signs, conjunctivitis, and immunosuppression. **Respiratory toxins** are not commonly seen, but the substrate is often the culprit in these instances.

Because the lungs are free-floating in the pleuroperitoneal cavity, there are a number of extra-respiratory disorders that may lead to respiratory signs in reptiles, particularly in squamates. These include anything that will push on the lungs and decrease lung volume such as **coelomic effusion, coelomic mass** (reproductive masses being particularly common in females), and **organomegaly** (hepatomegaly being common in bearded dragons because of hepatic lipidosis). Lastly, cardiac diseases are not common in reptiles other than snakes and may lead to respiratory compromise as well. In snakes, most cardiac diseases are bacterial endocarditis associated with bacterial pneumonia. Pulmonary edema consecutive to congestive heart failure is not well characterized in reptiles.

The Ten Most Important Facts

1. Respiratory diseases are frequently chronic in reptiles and they frequently present with non-specific signs such as anorexia and lethargy. Commonly seen clinical signs include nasal and oral bubbling or discharge and breathing pattern changes (especially increased respiratory efforts). Aquatic chelonians may display buoyancy issues.
2. Transtracheal lavage are used commonly in reptiles, especially in snakes and lizards. Fluid can be submitted for cytology, bacterial culture and sensitivity, fungal culture, or viral PCR.
3. For radiographic positioning, 3 views are needed in chelonians and the thoracic limbs of lizards should not overlap with the thorax. CT-scan is particularly useful in respiratory disorders of reptiles, especially in chelonians.
4. Different endoscopic approaches are available in reptiles to investigate respiratory diseases including tracheoscopy (but only possible in large reptiles), transcarapacial and prefemoral pulmonoscopy in chelonians, transcutaneous pulmonoscopy in squamates (especially in snakes), and routine coelioscopy.
5. Testudinid herpesviruses (Scutaviruses) may cause significant morbidity and mortality in Mediterranean tortoises. Species susceptibility varies but generally Hermann's tortoises are more susceptible than others and species mixing is not recommended. TeHV-3 is the most pathogenic of them all. Survivors are permanently infected.
6. Ferlavirus reptilis causes neurorespiratory signs in many snakes (particularly in viperidae and elapidae) and occasionaly in lizards and chelonians. The diagnosis is made by PCR on tracheal swab, transtracheal wash, or pulmonary biopsy.
7. Serptentoviruses are a common cause of respiratory infection in pythons. Diagnosis is made by PCR on tracheal swab, transtracheal wash, or pulmonary biopsy. Several species are implicated.
8. Primary or secondary (to a virus) bacterial pneumonia is also common in snakes, particularly in pythons. Gram negative bacteria are typically implicated. The treatment can be challenging and typically requires months of antibiotic therapy and improving the husbandry.
9. Mycoplasmosis of *Gopherus* tortoises is caused by *Mycoplasma agassizzi* and *Mycoplasma testudineum* and is a major cause of chronic rhinitis in these species. Diagnosis is made by PCR on a choanal swab or nasal flush. Treatment consists of long-term macrolide administration, but tortoises are seldom cleared of the bacteria and should be considered permanently infected.
10. Other respiratory diseases of note are tracheal chondroma of ball pythons, saprophytic fungi infection (especially *Purpureocillium lilaceum* and *Beauvaria bassiana*), and respiratory parasites (pentastomides in snakes, lung worms, and spirorchids termatodes in turtles).